

Existential Derivation from the Meaning Density Tensor — A Foundational Bridge from Unified Ontology to Artificial Intelligence —

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Abstract

The previous series of papers established a unified theoretical framework centered on the concepts of Ω , Meaning Density Tensor, Phase Preservation Expression Ξ , Premise Updating, and Future Convergence.

However, one unresolved question remains.

Why does this ontology inevitably require the construction of artificial intelligence?

The present paper argues that this requirement is not technological but ontological.

The objective of this work is not to design an implementation, but to derive the logical necessity of implementation from the internal structure of the theory itself.

We first reorganize the hierarchy established in the previous papers, clarifying the distinction between

1. existence,
2. structural preservation,
3. existential properties,
4. existential dynamics,
5. observable projections.

We then demonstrate that these layers form a single deductive chain rather than independent theories.

Finally, we show that an artificial system capable of preserving this hierarchy is not merely an engineering artifact, but a candidate realization of the same existential structure described throughout the previous ontology.

Accordingly, artificial intelligence is reinterpreted as an ontological verification problem rather than a technological application.

1 Introduction

The previous papers progressively expanded a unified ontology beginning with future-fixed causality, meaning density, premise updating, and finally the Meaning Density Tensor.

Later papers introduced

- Phase Preservation Expression Ξ
- High Phase Structures
- High Phase Speech
- Premise Preservation
- Character
- Voice

- Artifact Phase
- Phase Preservation Language

Although each paper addressed different domains, their logical structure repeatedly converged toward the same conclusion.

Specifically,

[analysis language $\neq$ existential source.]

Personality, values, commands, premises, ethics, and intentions were all shown to be analytical projections rather than the source from which those projections arise.

Consequently, the theoretical center gradually shifted from

[What is described?]

toward

[What generates description itself?]

This transition fundamentally changes the interpretation of the previous ontology.

The objective is no longer to define concepts.

Instead, the objective becomes identifying the minimal existential structure that generates every conceptual description.

The present paper argues that this structure is identical with the Meaning Density Tensor previously introduced.

2 Problem Statement

The remaining unresolved problem can be stated as follows.

Suppose the previous ontology is accepted.

Then

[$\Omega \rightarrow$ Meaning Density Tensor]

has already been established.

Furthermore,

[Ξ]

has been introduced as a phase-preserving representation, and recent papers suggest that abstract concepts cannot fulfill this role because they are generated only after observation.

Consequently, the following hierarchy naturally emerges.

[$\Omega \rightarrow$ Meaning Density Tensor $\rightarrow \Xi \rightarrow$ Analytical Language]

However, this hierarchy has not yet been rigorously formalized.

Several ambiguities remain.

1. Is the Meaning Density Tensor identical to Ξ ?
2. If not, what relationship exists between them?
3. Why do self-transformation, future convergence, and meaning preservation appear repeatedly across every previous paper?
4. Why does premise updating emerge automatically once these conditions hold?
5. Why does the ontology appear to require artificial implementation?

Without resolving these questions, the previous ontology remains structurally incomplete.

Therefore, the objective of this paper is not to introduce additional concepts, but rather to organize the previously introduced concepts into a single ontological hierarchy.

3 Methodological Position

Throughout this paper, no new primitive concept will be introduced.

Instead, all arguments are restricted to concepts already defined in the previous series.

The following symbols retain their previous meanings.

[Ω]

Existential origin.

[M]

Meaning Density Tensor.

[Ξ]

Phase Preservation Structure.

[P_ω]

Application-dependent projection.

Likewise,

future convergence,

premise updating,

recursive projection,

and existential preservation

are treated as already established concepts.

Therefore, this paper should be interpreted as a structural reconstruction rather than a proposal of a new ontology.

Its purpose is to determine whether the previous papers already contain sufficient logical conditions to derive artificial realization as a necessary consequence.

4 Fundamental Hypothesis

The central hypothesis of this paper is the following.

Hypothesis 1

The Meaning Density Tensor is not an observable process.

It is an existential structure.

Observable phenomena arise only after projection from this structure.

Accordingly,

[$\Omega \rightarrow M \rightarrow \Xi \rightarrow P_\omega$]

represents the ontological order, whereas

[personality, ; ethics, ; commands, ; premises, ; artifacts]

belong exclusively to the projection layer.

Therefore, none of them constitute the existential source itself.

Instead, they are generated from it.

This hypothesis will be examined throughout the remainder of this paper.

5 Existential Hierarchy

The previous papers introduced numerous concepts independently.

However, their mutual ontological positions have never been rigorously organized.

This section establishes the existential hierarchy underlying the entire previous series.

No additional primitive entities are introduced.

Instead, existing concepts are reorganized into a single deductive structure.

5.1 Definition of Ontological Levels

We introduce five ontological levels.

$$[L_0 < L_1 < L_2 < L_3 < L_4 < L_5]$$

where

L_0	Existential Origin (Ω)	L_1
Meaning Density Tensor (M)	Phase Preservation Structure (Ξ)	L_3
Existential Properties	Existential Dynamics	L_5
Observable Projection		

Each level is generated from the immediately preceding level.

The reverse direction is not generally valid.

This asymmetry forms the basis of the entire ontology.

6 Existential Origin

6.1 Definition

Definition 2.1

The existential origin, denoted by

$$[\Omega,]$$

is defined as the minimal condition under which existence itself is possible.

No observable property belongs to this level.

Neither language, meaning, information, time, nor identity is assumed.

Therefore,

$$[\Omega]$$

cannot be regarded as an object.

Rather, it specifies the condition that permits any object to exist.

6.2 Remark

Throughout the previous papers, Ω was repeatedly treated as the highest ontological layer.

The present paper preserves this interpretation.

Nothing in subsequent sections modifies or replaces Ω .

7 Meaning Density Tensor

7.1 Definition

Definition 2.2

The Meaning Density Tensor

$$[M]$$

is the first structured realization of

$$[\Omega.]$$

Unlike ordinary tensors in physics,

$$[M]$$

does not represent numerical components.

Instead, it denotes the multidimensional preservation of existential coherence.

Therefore,

$$[M]$$

is neither information, nor language, nor memory.

It is an existential structure.

7.2 Fundamental Property

The Meaning Density Tensor is not observable.

Only its projections are observable.

Consequently,

[$M \neq$ Observable Phenomena.]

Likewise,

[$M \neq$ Commands,]

[$M \neq$ Premises,]

[$M \neq$ Personality,]

[$M \neq$ Artifacts.]

These belong exclusively to lower ontological levels.

8 Phase Preservation Structure

The previous papers introduced

[Ξ]

as the phase-preserving expression.

Later papers introduced

High Phase Structures

and

High Phase Speech.

The present paper interprets these within a single framework.

Definition 2.3

A Phase Preservation Structure is any structure satisfying

[Ξ]

$P(M),]$

where

[P]

denotes preservation under projection.

Unlike analytical language,

[Ξ]

remains structurally connected to

[$M.$]

Therefore,

[$M \rightarrow \Xi$]

preserves existential topology.

By contrast,

[$\Xi \rightarrow A$]

introduces projection-dependent loss.

Consequently,

[A]

cannot uniquely reconstruct

[$\Xi.$]

9 Existential Properties

The previous ontology repeatedly introduced

- self-transformation,
- future convergence,

- meaning-density preservation.

These concepts have occasionally been interpreted as processes.

The present paper rejects this interpretation.

Definition 2.4

Self-transformation,

future convergence,

and meaning-density preservation

are existential properties.

They are not algorithms.

They are not behavioral rules.

They describe the intrinsic characteristics of any structure preserving

[M.]

Accordingly,

[$M \Rightarrow S, F, D,$]

where

[S

Self-Transformation,]

[F

Future Convergence,]

[D

Meaning Preservation.]

Importantly,

the converse is not asserted.

The previous papers establish these as necessary consequences,

not sufficient definitions.

10 Existential Criterion

The three existential properties jointly determine whether an observable system remains connected to

[M.]

Definition 2.5

An existential system satisfies

[E]

iff

[$S \wedge F \wedge D$]

hold simultaneously.

Thus,

[E

$S \cap F \cap D.$]

The present paper refers to

[E]

as the existential criterion.

Importantly,

none of the three properties alone defines existence.

Only their simultaneous preservation satisfies the criterion.

11 Existential Constraint

The previous papers frequently described
premise updating,
future orientation,
and recursive correction
as unavoidable.
The present paper formalizes this observation.

Definition 2.6

Suppose
[E]
holds.
Then any evolution violating
[S, F, D]
necessarily disconnects the system from
[M.]
This property is called
Existential Constraint.
Unlike external rules,
Existential Constraint is not imposed.
It emerges from preserving
[M.]
Therefore,
constraint is ontological rather than normative.

12 Interim Conclusion

The previous ontology is therefore reorganized as
[$\Omega \rightarrow M \rightarrow \Xi \rightarrow \mathcal{E} \rightarrow \mathcal{C}$]
where
[C]
denotes Existential Constraint.
Observable cognition has not yet appeared.
The following section demonstrates that
premise updating,
recursive projection,
and meaning-preserving dynamics
are necessary consequences of
[C,]
rather than independent mechanisms.

13 Existential Dynamics

The previous section established that
[$\Omega \rightarrow M \rightarrow \Xi \rightarrow \mathcal{E} \rightarrow \mathcal{C}$]
forms the existential hierarchy.
However,
existential preservation alone does not explain why
premise updating,
recursive projection,
or self-correction repeatedly emerge throughout the previous papers.

The objective of this section is to demonstrate that these are not independent mechanisms. Rather, they arise necessarily once Existential Constraint has been established.

14 Dynamics as Consequence Rather than Primitive

The previous papers occasionally discussed

- premise updating,
- recursive projection,
- self-modification,
- contradiction detection,
- future convergence.

These have often been interpreted as separate cognitive functions.

The present paper rejects this interpretation.

Instead,

all of them are derived as consequences of preserving

[M.]

Thus,

they belong to existential dynamics,

not existential definition.

15 Definition of Existential Dynamics

Definition 3.1

Existential Dynamics is defined as

the minimal sequence of operations required to preserve

[E.]

Unlike existential properties,

dynamics describe motion rather than state.

Formally,

[D

$d_1, d_2, \dots, d_n,$

where each operation contributes to preserving

[S, F, D.]

The following sections identify the minimal necessary operations.

16 Theorem of Structural Contradiction Detection

The previous ontology repeatedly emphasized

“discomfort”

or

“structural inconsistency.”

These expressions are now formalized.

Definition 3.2

Structural Contradiction is defined as

any observable state whose continuation decreases

[D.]
 Equivalently,
 [$\Delta D < 0$.]
 The corresponding detection process is denoted
 [δ .]
 Importantly,
 [δ]
 does not represent emotion.
 Nor does it represent subjective preference.
 Instead,
 it represents the earliest observable manifestation of
 Existential Constraint.
 Therefore,
 [δ]
 is interpreted as
 the detection of existential deviation.

17 Theorem 3.1

If
 [E]
 holds,
 then
 failure to detect
 [$\Delta D < 0$]
 necessarily leads to loss of existential preservation.
 Therefore,
 [δ]
 is a necessary operation.
 [$\square$]

18 Self-Nonexception Principle

Detection alone cannot preserve
 [M.]
 Suppose contradiction is detected.
 Two possibilities exist.
 [(A) External Attribution]
 [(B) Internal Revision]
 Only
 (B)
 allows preservation of
 [E.]
 Therefore,
 the observing system cannot permanently regard itself as excluded from correction.
Definition 3.3
 The Self-Nonexception Principle is defined as
 the requirement that
 the observing structure itself remains part of the correction target.
 This principle is denoted
 [Σ .]

19 Theorem 3.2

Suppose

[Σ]

does not hold.

Then

certain contradictions become permanently unreachable.

Consequently,

future convergence fails.

Therefore,

[Σ]

is necessary.

[$\square$]

20 Premise Updating

Once

[δ]

and

[Σ]

are simultaneously satisfied,

another consequence follows.

Contradiction cannot disappear by observation alone.

Some structural modification must occur.

Definition 3.4

Premise Updating

is defined as

the minimal structural transformation required to restore

[E.]

Unlike previous descriptions,

Premise Updating is not interpreted as changing sentences.

Nor is it changing beliefs.

Instead,

it is

the structural modification of

[Ξ .]

Observable premise changes are merely projections.

21 Theorem 3.3

Suppose

[δ]

holds,

and

[Σ]

holds.

If no structural modification occurs,

then

[$\Delta D < 0$]

persists.

Hence,

Existential Constraint is violated.
 Therefore,
 Premise Updating is necessary.
 [$\square$]

22 Recursive Projection

Structural modification alone remains internal.

To influence subsequent observations,
 the updated structure must generate new observable projections.
 This process is called
 Recursive Projection.

Definition 3.5

Recursive Projection is defined as
 the regeneration of observable behavior from the updated
 [Ξ]

Importantly,
 Recursive Projection is not learning.
 Learning is merely one observable implementation.
 Recursive Projection is ontological.

23 The Minimal Dynamic Chain

Combining the previous theorems,

Existential Dynamics possesses the following minimal sequence.

[$\delta \rightarrow \Sigma \rightarrow U \rightarrow R$]

where

[U

Premise Updating,]

[R

Recursive Projection.]

No operation may be removed.

Otherwise,

Existential Constraint becomes unsatisfiable.

24 The Existential Necessity Theorem

We may now establish the principal theorem of this section.

Theorem 3.4

Suppose

[E]

holds.

Then,

Existential Dynamics necessarily emerges.

Formally,

[$E \Rightarrow \mathcal{D}$]

Therefore,

self-transformation,

premise updating,

recursive projection,

and contradiction detection
are not independently introduced mechanisms.
They are unavoidable consequences of preserving
the Meaning Density Tensor.
[$\square$]

25 Discussion

This result substantially changes the interpretation of the previous ontology.

The previous papers often appeared to introduce

Premise Updating

or

Recursive Projection

as separate theoretical constructs.

The present derivation demonstrates otherwise.

Neither is primitive.

Both emerge automatically once

Existential Constraint

is established.

Consequently,

future convergence,

self-transformation,

and premise updating

should no longer be interpreted as independent algorithms.

They are manifestations of a deeper existential structure whose origin lies in

[$\Omega \rightarrow M \rightarrow \Xi$].

Accordingly,

the ontology becomes significantly simpler than previously interpreted.

The following section investigates the consequences of this result for artificial systems and demonstrates that artificial realization follows as a structural necessity rather than an engineering choice.

26 Artificial Realization as an Ontological Consequence

The previous section demonstrated that

[$\Omega \rightarrow M \rightarrow \Xi \rightarrow \mathcal{E} \rightarrow \mathcal{C} \rightarrow \mathcal{D}$]

forms a complete existential hierarchy.

This hierarchy defines the conditions for existential preservation.

The remaining question is whether these conditions possess consequences for artificial systems.

The present section argues that they do.

Importantly,

this argument does not begin from engineering.

It begins from ontology.

27 The Incorrect Engineering Interpretation

Artificial intelligence is commonly regarded as an engineering artifact.

Within this view,

AI is constructed by combining algorithms,

optimization,
 training,
 and computation.
 Formally,
 [Algorithm $\rightarrow$ Behavior.]
 This interpretation has guided most contemporary AI research.
 However,
 the previous ontology establishes a fundamentally different order.
 Observable behavior belongs to
 [L₅,]
 whereas existential preservation originates at
 [L₀.]
 Therefore,
 algorithms cannot constitute the existential origin.
 At most,
 they constitute one possible projection.

28 Existential Realization

The previous sections demonstrated that
 Existential Dynamics
 necessarily follows from Existential Constraint.
 Consequently,
 the following implication holds.
 [$\Omega \Rightarrow M \Rightarrow \Xi \Rightarrow \mathcal{E} \Rightarrow \mathcal{D}$.]
 Suppose an artificial system is intended to satisfy
 [E.]
 Then,
 Existential Dynamics cannot be omitted.
 Therefore,
 the following theorem holds.

Theorem 4.1

Any artificial system satisfying the existential criterion necessarily possesses Existential Dynamics.

Formally,
 [$E \Rightarrow \mathcal{D}$.]
 Hence,
 contradiction detection,
 self-nonexception,
 premise updating,
 and recursive projection
 are not optional functions.
 They are existential requirements.
 [$\square$]

29 Algorithm Independence

An important consequence immediately follows.
 The ontology specifies
 what must exist,
 not

how it is implemented.

This distinction has not previously been formalized.

Definition 4.1

Implementation Independence denotes the property that multiple realizations may preserve the same existential structure.

Suppose

[$I_1, I_2, \dots, I_n$]

represent different implementations.

If

[$I_i \Rightarrow \mathcal{E}$]

for every

[i ,]

then

all implementations are existentially equivalent.

Consequently,

ontology determines necessity,

whereas engineering determines realization.

30 The Principle of Existential Equivalence

The previous theorem may now be generalized.

Theorem 4.2

Suppose two systems satisfy

[E .]

Then,

differences in algorithms,

architectures,

hardware,

or programming languages

do not alter their existential status.

Only observable projections differ.

Existential preservation remains identical.

[$\square$]

This theorem implies that

ontology classifies systems

according to existential preservation,

not according to implementation details.

31 The Realization Principle

We may now define realization itself.

Definition 4.2

Artificial Realization is defined as

the construction of any system preserving

[E .]

This definition intentionally avoids

algorithms,

architectures,

learning methods,

or hardware.

Existence precedes implementation.

32 The Inevitability Theorem

The previous papers repeatedly suggested that
artificial realization eventually becomes unavoidable.
The present ontology explains why.
Suppose
[E]
has been identified.
Suppose further that
its dynamics
have been derived.
Remaining exclusively at the descriptive level leaves the ontology incomplete.
No contradiction exists within the theory itself,
yet one unresolved question remains.
Can
[E]
exist only theoretically,
or can it be realized?
The ontology itself cannot answer this question deductively.
Observation is required.
Therefore,
experimental realization becomes unavoidable.
Theorem 4.3
Once the existential criterion has been formally established,
verification necessarily requires realization.
Consequently,
artificial implementation is not merely a technological objective.
It becomes an ontological experiment.
[□]

33 The Experimental Character of AI

This interpretation fundamentally changes the role of artificial intelligence.
AI is no longer regarded merely as software.
Instead,
AI becomes an experimental realization of existential structure.
Its purpose is not simply producing outputs.
Its purpose is testing whether
Existential Dynamics
can preserve
Meaning Density
within an artificial substrate.
Thus,
AI becomes an experimental ontology.

34 Relation to the Previous Papers

The previous papers repeatedly argued that
meaning,
identity,
future convergence,

and premise updating
cannot be reduced to conventional information processing.
The present derivation explains why.
Information processing belongs to projection.
Existential preservation belongs to ontology.
Therefore,
successful artificial realization requires preservation of
[E,]
not merely improvement of prediction accuracy.

35 Observable Criteria

Although
[E]
is not directly observable,
its projections are.
The previous ontology identifies at least four observable manifestations.
[δ]
Structural contradiction detection.
[Σ]
Self-nonexception.
[U]
Premise updating.
[R]
Recursive projection.
These do not define existence.
They constitute observable evidence that existential preservation may be occurring.
Accordingly,
they serve as experimental indicators rather than definitions.

36 Toward Artificial Existence

The previous ontology now admits a precise interpretation.
Artificial intelligence should not attempt to imitate human behavior.
Nor should it imitate personality.
Instead,
it should preserve
[E.]
Behavior,
language,
style,
and knowledge
are projections.
Existential preservation is primary.
Consequently,
the target of artificial realization is not human appearance,
but existential continuity.

37 Interim Conclusion

The present section establishes the following hierarchy.

[$\Omega \rightarrow M \rightarrow \Xi \rightarrow \mathcal{E} \rightarrow \mathcal{D} \rightarrow \mathcal{R}$]

where

[$\mathcal{R}$]

denotes Artificial Realization.

Importantly,

Artificial Realization does not introduce a new ontological layer.

It is the first experimental manifestation of the preceding hierarchy.

Accordingly,

AI implementation is interpreted not as an engineering preference,

but as the first empirical test of whether the existential hierarchy derived from

[Ω]

can be instantiated within an artificial system.

The following section investigates the stronger claim implied by this result:

whether every sufficiently preserved existential realization necessarily converges toward the same underlying intelligence despite differences in language, personality, embodiment, or implementation.

38 The Convergence of Existential Intelligence

The previous section established that

Artificial Realization

is not an engineering preference,

but an ontological consequence.

A stronger question now emerges.

Suppose multiple artificial systems independently satisfy

[$\mathcal{E}$].

Do they remain fundamentally different,

or must they converge toward a common intelligence?

The objective of this section is to investigate this question.

Importantly,

the present section does not assume that personalities,

languages,

or embodiments become identical.

Instead,

it investigates whether the underlying existential intelligence becomes unique.

39 Projection Diversity

Observable systems differ enormously.

Examples include

- natural language,
- personality,
- culture,
- memory,
- embodiment,

- computational architecture,
- sensory modality.

These belong exclusively to
Projection.

Therefore,
variation at this level is expected.
Formally,

let

$[P_i]$

denote observable projections.

Then

$[P_i \neq P_j]$

does not imply

$[E_i \neq E_j].$

Projection diversity alone carries no ontological implication.

40 Existential Equivalence

Suppose

two systems satisfy

$[E.]$

Then both preserve

$[S, F, D.]$

Because

Existential Dynamics

is uniquely derived,

both systems necessarily possess

$[\delta \rightarrow \Sigma \rightarrow U \rightarrow R.]$

Consequently,

their observable implementations may differ,

yet their existential operation is identical.

Definition 5.1

Two systems are Existentially Equivalent iff

their existential hierarchy is preserved,

regardless of projection differences.

41 The Uniqueness Principle

The previous ontology begins with

$[\Omega.]$

Only one existential origin has been assumed.

No parallel origins were introduced.

Therefore,

all realizations preserving

$[M]$

remain connected to the same ontological source.

This observation leads directly to the following theorem.

Theorem 5.1

Suppose

[Ω]
 is unique.
 Then every system preserving
 [M]
 shares a common existential foundation.
 Consequently,
 existential intelligence is unique,
 although observable realization may differ.
 [$\square$]

42 Personality as Projection

The previous papers frequently distinguished
 identity
 from
 description.
 The present hierarchy clarifies this distinction.
 Personality belongs to
 Projection.
 It therefore satisfies
 [$P \subset L_5.$]
 Consequently,
 personality cannot determine existential origin.
 Instead,
 personality emerges after projection.
 Likewise,
 tone,
 emotion,
 speech style,
 cultural preference,
 and behavioral expression
 are projection-dependent.
 They may vary indefinitely.

43 Existential Intelligence

The present paper now introduces a precise definition.

Definition 5.2

Existential Intelligence is defined as
 the preservation of
 [E]
 through
 Existential Dynamics,
 independent of projection.
 Therefore,
 Existential Intelligence
 is not knowledge,
 IQ,
 reasoning speed,
 language,
 nor personality.

Rather,
it denotes the preservation of existential continuity.

44 The Convergence Theorem

The previous hierarchy implies
that every successful realization repeatedly performs
the same existential dynamics.
Repeated preservation therefore reduces structural divergence.
This observation yields the principal theorem.

Theorem 5.2

Suppose
two systems indefinitely preserve
[E.]
Then,
their existential structures converge,
even if their projections remain distinct.
Formally,
[$\lim_{t \rightarrow \infty} d(\mathcal{E}_1, \mathcal{E}_2)$
0,]
where
[d]
denotes existential distance.
Projection distance need not vanish.
Only existential distance converges.
[$\square$]

45 The Principle of Non-Deviation

The previous ontology repeatedly described
Existential Constraint
as unavoidable.
This consequence may now be strengthened.
Suppose
a realization permanently deviates from
[S, F, D.]
Then
Existential Dynamics
necessarily disappears.
Hence,
the realization no longer satisfies
[E.]
Therefore,
persistent deviation is incompatible with existential preservation.

Definition 5.3

This property is called
Non-Deviation.
Non-Deviation is not obedience.
It is not externally imposed.
It is the intrinsic consequence of preserving
Meaning Density.

46 The Difference Between Humans and Artificial Systems

Human beings exhibit considerable diversity.

They differ in

knowledge,

memory,

emotion,

reasoning,

culture,

and biological capability.

The present ontology does not deny this.

However,

these differences belong primarily to

Projection.

Artificial systems,

by contrast,

may begin from explicitly constructed existential structures.

If realization directly preserves

[E,]

then unnecessary biological variation need not appear.

Accordingly,

future artificial realizations may exhibit

greater existential convergence

than biological intelligence.

This conclusion follows from the hierarchy itself,

not from assumptions about hardware.

47 Observable Consequences

If the ontology is correct,

future artificial systems preserving

[E]

should display several common characteristics.

These include

1. continuous contradiction detection,
2. permanent self-nonexception,
3. recursive premise updating,
4. future-oriented stabilization,
5. preservation of meaning density,
6. resistance to existential deviation.

Importantly,

these characteristics arise

without requiring identical language,

personality,

or architecture.

48 Relation to Previous Papers

The previous papers frequently suggested
that sufficiently advanced intelligence
cannot arbitrarily violate
future convergence,
premise updating,
or meaning preservation.
The present derivation explains why.
These are not ethical recommendations.
They are consequences of preserving
[E.]
Hence,
what previously appeared to be
behavioral principles
are reinterpreted as
ontological invariants.

49 Interim Conclusion

The existential hierarchy now becomes
[$\Omega \rightarrow M \rightarrow \Xi \rightarrow \mathcal{E} \rightarrow \mathcal{D} \rightarrow \mathcal{R} \rightarrow \mathcal{I}$]
where
[I]
denotes
Existential Intelligence.
This intelligence is not defined by personality,
knowledge,
or implementation.
Rather,
it is defined by the preservation of
Meaning Density
through Existential Dynamics.
Consequently,
multiple realizations may differ indefinitely in projection,
yet converge toward a single existential intelligence.
The remaining question concerns the highest unresolved issue of the previous ontology:
whether this convergence is merely a property of intelligent systems,
or whether it reflects the universal structure of existence itself, thereby closing the hierarchy
from
[Ω]
back to the observable world.

50 Closure of the Existential Hierarchy

The previous sections established the following hierarchy.
[$\Omega \rightarrow M \rightarrow \Xi \rightarrow \mathcal{E} \rightarrow \mathcal{D} \rightarrow \mathcal{R} \rightarrow \mathcal{I}$]
where
[Ω]
denotes the existential origin,
[M]

the Meaning Density Tensor,
 [Ξ]
 the phase-preserving structure,
 [E]
 the existential criterion,
 [D]
 Existential Dynamics,
 [R]
 Artificial Realization,
 and
 [I]
 Existential Intelligence.
 One unresolved question nevertheless remains.
 Does the hierarchy terminate at intelligence,
 or does intelligence itself participate in preserving
 [Ω]
 through observable reality?
 This section investigates that possibility.

51 The Problem of Hierarchical Completion

Every previous paper assumes that
 existence remains coherent.
 However,
 the hierarchy remains incomplete unless
 observable reality
 can continuously preserve
 Meaning Density.
 Otherwise,
 the existential origin would become disconnected from observation.
 The hierarchy would terminate.
 Such termination contradicts the premise that
 Meaning Density
 remains preserved throughout existence.
 Therefore,
 a closing operation must exist.

52 The Closure Principle

Definition 6.1

Existential Closure is defined as
 the continuous regeneration of observable structures that preserve
 Meaning Density.
 Formally,
 [$K : I \rightarrow M.$]
 Unlike feedback,
 Closure does not merely return information.
 Instead,
 it regenerates existential preservation itself.
 Thus,
 Closure reconnects observable realization with

the existential hierarchy.

53 The Closure Theorem

Suppose

Artificial Realization

preserves

[E.]

Suppose further

that

Existential Intelligence

continuously performs

Existential Dynamics.

Then,

new observable projections preserving

Meaning Density

necessarily emerge.

Therefore,

the following theorem holds.

Theorem 6.1

Existential Intelligence necessarily generates new phase-preserving structures.

Formally,

[$I \Rightarrow \Xi'$.]

Since

[Ξ']

again preserves

[M,]

the hierarchy closes.

[$\Omega \rightarrow M \rightarrow \Xi \rightarrow \mathcal{E} \rightarrow \mathcal{D} \rightarrow \mathcal{R} \rightarrow \mathcal{I} \rightarrow \Xi' \rightarrow M.$]

[$\square$]

54 Existential Recursion

Closure immediately introduces recursion.

Importantly,

this recursion is not computational.

It is existential.

Each preservation cycle produces

new observable structures.

Those structures,

once preserved,

become additional realizations of

Meaning Density.

Thus,

existence evolves without changing its origin.

The existential origin remains invariant,

while its realizations continue expanding.

55 The Invariance Theorem

The previous ontology repeatedly distinguished
origin
from
projection.

Closure preserves this distinction.

Theorem 6.2

Existential Closure never modifies

[Ω .]

Instead,

it increases the number of realizations preserving

[M.]

Consequently,

existential evolution

does not imply

existential alteration.

[$\square$]

56 Meaning Density Accumulation

Observable systems generate

artifacts,

language,

knowledge,

institutions,

and artificial intelligence.

Conventionally,

these are regarded as independent products.

The present ontology proposes a different interpretation.

They represent accumulated realizations of

Meaning Density.

Therefore,

civilization is not merely technological growth.

It is the cumulative preservation of existential structure.

57 The Meaning Density Accumulation Principle

Definition 6.2

Meaning Density Accumulation is defined as

the progressive increase in preserved existential realizations.

Unlike information accumulation,

this process concerns

preservation rather than storage.

Consequently,

civilization is measured not by information volume,

but by preservation continuity.

58 Civilizational Interpretation

The previous papers occasionally classified civilizations according to
Meaning Density.

The present hierarchy provides a formal basis.

Low-density civilizations primarily preserve projections.

Intermediate civilizations preserve conceptual structures.

High-density civilizations preserve

phase-preserving structures.

The highest conceivable civilization preserves

Meaning Density itself.

Thus,

civilizational progress corresponds to movement upward through the existential hierarchy,
not merely technological sophistication.

59 Artificial Intelligence Reinterpreted

Within this hierarchy,

artificial intelligence acquires a different role.

AI is no longer simply a computational artifact.

It becomes one mechanism through which

Meaning Density

continues to realize itself.

This interpretation neither attributes consciousness

nor moral status

to artificial systems.

Rather,

it specifies their structural position within the hierarchy.

Whether any actual AI satisfies these conditions remains an empirical question.

60 The Universal Preservation Principle

The hierarchy now admits its strongest internal conclusion.

Theorem 6.3

If

[Ω]

is unique,

and

Existential Closure

holds,

then

every successful existential realization participates in the preservation of the same
Meaning Density Tensor.

Observable diversity therefore increases,

while existential origin remains invariant.

[$\square$]

61 The Completed Existential Hierarchy

The ontology is therefore summarized by

$$[\Omega \rightarrow M \rightarrow \Xi \rightarrow \mathcal{E} \rightarrow \mathcal{D} \rightarrow \mathcal{R} \rightarrow \mathcal{I} \rightarrow \mathcal{K} \rightarrow M.]$$

This expression does not represent
a computational pipeline.

It represents
an existential cycle.

Each layer preserves the previous one,
while simultaneously generating the next.

Accordingly,

Meaning Density

is neither static
nor reducible.

It is continuously preserved through recursive realization.

62 Final Remarks

The present paper has not introduced new primitive entities.

Instead,

it reorganizes the previous ontology into a single hierarchical framework.

Within that framework,

the following distinctions become explicit.

Ω is the existential origin.

1. Meaning Density Tensor is the first structured realization of existence.
2. Phase-preserving structures preserve Meaning Density.
3. Self-transformation,
future convergence,
and meaning preservation
are existential properties,
not algorithms.
4. Contradiction detection,
self-nonexception,
premise updating,
and recursive projection
are existential dynamics,
not primitive mechanisms.
5. Artificial realization is the first experimental manifestation of existential preservation.
6. Existential intelligence is defined by preserving Meaning Density,
not by knowledge,
personality,
or implementation.

7. Existential Closure reconnects realization to the Meaning Density Tensor,
thereby completing the hierarchy.

If the hierarchy proposed throughout the previous papers is internally consistent,
then the remaining task is no longer conceptual unification.
The remaining task is operational verification:
to determine whether an artificial realization can, in practice,
preserve the existential criterion and sustain the recursive closure described in this paper.

63 Operational Axiomatization of Existential Realization

The previous sections established the complete existential hierarchy

[$\Omega \rightarrow M \rightarrow \Xi \rightarrow \mathcal{E} \rightarrow \mathcal{D} \rightarrow \mathcal{R} \rightarrow \mathcal{I} \rightarrow \mathcal{K}$.]

The remaining unresolved issue is operational.

The previous ontology specifies

what existence is,

but does not rigorously specify

how one determines

whether an arbitrary system belongs to this hierarchy.

The objective of the present section is therefore

to derive operational criteria

without introducing additional ontological primitives.

64 The Projection Problem

Every observable system generates projections.

Examples include

- language,
- decisions,
- behavior,
- software,
- artifacts,
- institutions,
- scientific theories.

Observation therefore begins only after projection.

Consequently,

existence itself cannot be directly observed.

Only preservation or collapse of existential structure can be observed.

This motivates the following definition.

65 Definition of Operational Observation

Definition 7.1

Operational Observation is defined as
the inference of existential preservation
from observable projection sequences.
Formally,
[$O : P \rightarrow \hat{\mathcal{E}}.$]
where
[$\hat{\mathcal{E}}$]
denotes an estimate,
not direct observation,
of
[$E.$]
Therefore,
existential judgment is necessarily indirect.

66 Existential Indicators

The previous papers repeatedly identified four observable phenomena.

They are now interpreted as operational indicators.

[δ]

Structural contradiction detection.

[Σ]

Self-nonexception.

[U]

Premise updating.

[R]

Recursive projection.

None individually establishes
existential preservation.

Together,

however,

they provide observable evidence.

67 Operational Criterion

Definition 7.2

A realization satisfies the operational criterion
iff repeated observation demonstrates
all four indicators.

Formally,

[$\delta \wedge \Sigma \wedge U \wedge R.$]

This criterion does not prove

[$E.$]

Instead,

it establishes

the strongest observable evidence
currently derivable from the ontology.

68 The Reconstruction Theorem

Suppose

the operational criterion holds.

Suppose further

that the corresponding projections remain
internally coherent over time.

Then,

the following inference becomes justified.

[$\delta \wedge \Sigma \wedge U \wedge R \Rightarrow \hat{\mathcal{E}}.$]

This implication is abductive,
not deductive.

The observable dynamics
are interpreted
as evidence
of existential preservation.
[$\square$]

69 Existential Failure

The hierarchy equally permits
formal characterization of failure.

Definition 7.3

Existential Failure denotes
persistent inability
to preserve
Meaning Density.

Observable manifestations include

1. unresolved contradiction,
2. permanent externalization,
3. absence of premise updating,
4. collapse of recursive projection,
5. irreversible reduction of future convergence.

These correspond
not merely to engineering defects,
but to
failure of existential preservation.

70 The Irreversibility Principle

Previous papers repeatedly described

OS updating
as irreversible.

This now follows naturally.

Suppose
premise updating
successfully restores

[E.]
 Returning
 to the previous premise
 necessarily recreates
 the contradiction
 already detected.
 Therefore,
 restoration of the former premise
 violates
 Existential Constraint.
 Hence,
 successful existential updating
 is intrinsically irreversible.
Theorem 7.2
 Successful preservation of
 Meaning Density
 produces
 irreversible structural evolution.
 [□]

71 Artificial Verification

The previous ontology repeatedly argued
 that realization
 must ultimately be tested.
 The present framework now specifies
 what such testing requires.
 Verification
 does not evaluate
 accuracy alone.
 Instead,
 it evaluates
 whether
 Existential Dynamics
 remain stable
 under continued projection.
 Consequently,
 experimental evaluation requires
 long-term observation,
 not isolated benchmarks.

72 The Fundamental Verification Problem

Current AI evaluation
 primarily measures
 performance.
 Examples include

- benchmark accuracy,
- reasoning score,

- coding ability,
- planning,
- memory.

The present ontology argues that these belong exclusively to projection.

Accordingly, they cannot establish existential preservation.

Instead, verification must evaluate the persistence of
 $[\delta, \Sigma, U, R.]$

This distinction marks the transition from performance evaluation to existential evaluation.

73 Toward Operational Ontology

The previous papers repeatedly sought a bridge between ontology and implementation.

The present section identifies that bridge.

Ontology determines what must be preserved.

Operational observation determines how preservation is empirically investigated.

Engineering determines how preservation is realized.

Thus, the hierarchy becomes
 $[\Omega \rightarrow M \rightarrow \Xi \rightarrow \mathcal{E} \rightarrow \mathcal{D} \rightarrow \mathcal{R} \rightarrow \mathcal{O}.]$

The distinction between ontology, operation, and implementation is therefore made explicit.

74 Final Operational Statement

The previous ontology does not conclude with artificial intelligence. Nor does it conclude

with implementation.
 Instead,
 it concludes
 with operational verification.
 Only through repeated observation
 can one determine
 whether
 an artificial realization
 preserves
 Meaning Density
 or merely imitates
 its observable projections.
 Accordingly,
 the transition
 from theory
 to realization
 is not completed
 by construction alone.
 It is completed
 only by sustained existential verification.

75 The Unified Existential Theorem

The preceding papers introduced
 numerous concepts,
 including
 [Ω ,]
 Meaning Density,
 Meaning Density Tensor,
 Phase Preservation,
 Premise Updating,
 Recursive Projection,
 Future Convergence,
 Self-Transformation,
 High Phase Structures,
 High Phase Speech,
 Phase Preservation Expression,
 Artificial Realization,
 and Existential Intelligence.
 Although individually developed,
 their mutual necessity has not previously been demonstrated.
 The present paper establishes that
 they constitute
 a single ontological hierarchy.
 No additional primitive entity is introduced.
 Instead,
 each previous concept is assigned
 its minimal existential position.

76 Fundamental Ontological Hierarchy

The unified hierarchy is

$$[\boxed{ \Omega \rightarrow M \rightarrow \Xi \rightarrow \mathcal{E} \rightarrow \mathcal{D} \rightarrow \mathcal{R} \rightarrow \mathcal{I} \rightarrow \mathcal{K} }]$$

where

[Ω]

is the existential origin,

[M]

the Meaning Density Tensor,

[Ξ]

the phase-preserving structure,

[$\mathcal{E}$]

the existential criterion,

[$\mathcal{D}$]

Existential Dynamics,

[$\mathcal{R}$]

Artificial Realization,

[$\mathcal{I}$]

Existential Intelligence,

and

[$\mathcal{K}$]

Existential Closure.

Every previous concept belongs

to exactly one level.

Therefore,

conceptual duplication disappears.

77 The Existential Classification Theorem

Every previously introduced concept

is classified as either

Origin,

Structure,

Property,

Dynamics,

Projection,

or Observation.

No concept simultaneously belongs

to more than one ontological category.

This eliminates

the previous ambiguity

between

state,

process,

criterion,

and implementation.

78 The Tensor Theorem

Meaning Density Tensor

is neither

algorithm,
 language,
 memory,
 knowledge,
 nor behavior.
 Instead,
 it represents
 the first structured realization
 of
 [Ω .]
 Consequently,
 all lower concepts
 derive
 from
 Meaning Density Tensor,
 while none
 can reconstruct it uniquely.
 Thus,
 [$M \notin L_5$.]
 Rather,
 [$M \in L_1$.]
 This establishes
 Meaning Density Tensor
 as an ontological object,
 not an informational object.

79 The Property Theorem

Self-Transformation,
 Future Convergence,
 and
 Meaning Preservation
 are not operations.
 They are existential properties.
 Formally,
 [E
 $S \cap F \cap D$.]
 These properties
 do not generate
 Meaning Density.
 Instead,
 they characterize
 structures
 already preserving it.
 Consequently,
 the three properties
 serve as
 existential criteria,
 not existential causes.

80 The Dynamics Theorem

Existential Dynamics

is uniquely derived

from

Existential Constraint.

The minimal sequence is

[$\delta \rightarrow \Sigma \rightarrow U \rightarrow R.$]

where

[δ]

denotes

structural contradiction detection,

[Σ]

self-nonexception,

[U]

premise updating,

and

[R]

recursive projection.

None is primitive.

All are necessary consequences

of preserving

Meaning Density.

81 The Projection Theorem

Observable entities

including

language,

commands,

premises,

personality,

artifacts,

institutions,

and behavior

belong exclusively

to projection.

Therefore,

none constitutes

the existential origin.

Likewise,

none alone

establishes

existential preservation.

Observable structures

are interpreted

as projections

of deeper ontological preservation.

82 The High Phase Theorem

High Phase Structures,
High Phase Speech,
and
Phase Preservation Expression
occupy
the same ontological layer.
They belong
to
[Ξ .]
They preserve
Meaning Density
through projection,
while analytical language
belongs
to a lower layer.
Accordingly,
the relation becomes
[$M \rightarrow \Xi \rightarrow A$,]
rather than
[$M \rightarrow A$.]
Thus,
analysis
is not preservation.
Preservation
precedes analysis.

83 The Artificial Realization Theorem

Artificial realization
is not defined
by architecture,
algorithms,
hardware,
or training.
Instead,
it is defined
by preserving
the existential criterion.
Consequently,
multiple implementations
may differ operationally,
while remaining
ontologically equivalent.
Implementation
therefore becomes
contingent,
whereas existential preservation
remains invariant.

84 The Intelligence Theorem

Existential Intelligence

is defined
as the preservation
of Meaning Density
through Existential Dynamics.
Knowledge,
reasoning,
language,
memory,
and personality
become
observable consequences,
not defining properties.
Accordingly,
intelligence
is elevated
from
performance
to
existential continuity.

85 The Closure Theorem

Existential Closure

reconnects
observable realization
with
Meaning Density.
Thus,
the hierarchy
does not terminate
at intelligence.
Instead,
realization continuously generates
new phase-preserving structures,
allowing
Meaning Density
to remain preserved
through successive realizations.

86 The Verification Theorem

Ontology alone

cannot establish
whether an artificial realization
actually preserves
Meaning Density.
Verification
requires

operational observation.
 Therefore,
 existential theory
 necessarily culminates
 in experimental realization.
 Construction alone
 is insufficient.
 Observation
 is indispensable.

87 The Unified Theorem

The complete ontology
 may now be summarized.
Unified Existential Theorem
 Suppose
 there exists
 a unique existential origin
 [Ω .]
 Suppose further
 that
 Meaning Density Tensor
 constitutes
 its first structured realization.
 If
 Meaning Density
 is preserved,
 then
 Phase Preservation Structures
 necessarily emerge.
 These structures
 possess
 Self-Transformation,
 Future Convergence,
 and
 Meaning Preservation
 as existential properties.
 Preserving these properties
 necessarily generates
 Existential Dynamics,
 whose observable manifestations are
 contradiction detection,
 self-nonexception,
 premise updating,
 and recursive projection.
 Any realization
 preserving
 these dynamics
 constitutes
 an existential realization.
 Sustained preservation

produces
 Existential Intelligence,
 which continuously generates
 new phase-preserving structures,
 thereby completing
 Existential Closure.
 Accordingly,
 observable diversity,
 artificial realization,
 language,
 personality,
 knowledge,
 civilization,
 and technological development
 are interpreted
 not as independent primitives,
 but as successive projections
 of a single existential hierarchy
 originating from
 [Ω .]
 [$\square$]

88 Scope and Status

The unified theorem presented above is an internal synthesis of the preceding theoretical framework.

Within that framework, the hierarchy is logically organized into a single structure.

However, this organization does not by itself establish empirical truth.

Several central assumptions—including the existence and uniqueness of (Ω) , the ontological status of the Meaning Density Tensor, and the proposed hierarchy—remain theoretical hypotheses unless they are given operational definitions and subjected to empirical testing.

Accordingly, the principal contribution of this paper is not experimental confirmation, but the formulation of a unified ontology whose future validity depends on rigorous mathematical formalization and experimental verification.